



ROYAL GOVERNMENT OF BHUTAN
MINISTRY OF ENERGY AND NATURAL RESOURCES
Department Of Forest and Park Services
Mongar Forest Division
Lingmethang Forest Management Unit



Cross-cutting Management Regime (NWFPs) **for** **Broksar Shingmen Thuenken Detshen** **(2nd Revision)**



Saling Gewog, Mongar Dzongkhag
Plan Period: November 1, 2024 – October 31, 2034

APPROVAL SHEET

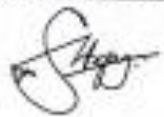
Approval Sheet

Name of NWFP Management and Marketing Group	Broksar Shingmen Thuenken Detshen
Village(s)	Broksar
Gewog	Saling
Dzongkhag/Dungkhag	Mongar
No. of households	124
Area (ha)	2533.11 ha (cane and bamboo block-712.87ha, other NWFP block-1804.723ha)
Plan period	November 2024 – October 2034

Prepared and submitted for Approval

	Chairperson of Management and Marketing Group	Gup of the Gewog	Head FRPM Section of Division/Park	Chief Forestry Officer of Division/Park
Signature:				
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Reviewed and recommended for approval

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Signature		
Name	Kezang Wangmo	Sonam Tobgay
Date	04/11/2024	04/11/2024

Approved by:



(DIRECTOR)

Department of Forests and Park Services, MoENR

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PART A: NWFP Management and Marketing Plan

1. Background

The *Broksar Shingmen Thuenken Detschen* is a cross-cutting management area identified for the management and marketing of Non-Wood Forest Products (NWFP) by group members formed for Broksar village. The village is located in the southern part of Saling Gewog, Mongar Dzongkhag, at an altitude of 1500 meters above sea level. The village is home to 24 households, and the primary dialect spoken is *Khengkha*. The forest, predominantly a cool broad-leaved forest, plays a critical role in the community's livelihood. The community relies on timber, NWFPs, livestock, and agriculture for income. Traditionally, farmers weave baskets from canes (*Plectocomia himalayana* and *Calamus acanthospathus*) for domestic use. The average household income varies annually, influenced by the sale of agricultural and forest products. The primary NWFPs collected include various cane species, utilized both for household consumption and sale to external buyers.

The community possesses traditional knowledge and skills related to the management and utilization of NWFPs. The increase in households and extensive resource collection by neighbouring Gewogs have led to resource depletion. This situation underscores the necessity for sustainable management practices. The group was formed to foster collective responsibility, protect resources, and ensure sustainable use. The Broksar Shingmen Thuenken Detschen was formed to manage, collect, and market NWFPs sustainably, particularly cane species, within the State Reserved Forest areas under their traditional rights. The group aims to generate significant income for its members through these.

The specific boundaries for NWFP collection are Raowbi and Trangla to the north, Khengpalaptsa, Jainla, and Ribalingbu to the east, lower Yarila and Shayrila to the south, and lower Trangla and upper Yarila to the west. If this document is a revision of the previous plan, it is essential to highlight past achievements and justify the need for the revision. This may include improved resource management outcomes, increased community income, and the necessity to address new challenges or opportunities in NWFP management. The revision aims to refine strategies to ensure ongoing sustainability and enhanced benefits for the community.

The NWFP management group was formed in the year 2019 with support from SFED fund and the plan got expired in December 2022. During the three years management plan period to ensure sustainable management of cane resources the group members harvested the canes based on the order received from the buyers through the Divisional Forest Office and strictly followed as per the AHL prescribed in the plan. The group members harvested canes comparatively low as per the AHL depicted in the below table.

Table 1. Annual Harvesting Limit during the 1st Plan

Year	AHL (Backloads)		Total AHL	Qty harvested (Backloads)	Remarks
	Block I (Thonphu I)	Block II (Thonphu II)			
2020	217	259	476	91	

2021	217	259	476	273
2022	217	259	476	285
Total	651	777	1428	649

All the group members managed and protected the whole NWFP management area. The members were allowed to harvest one backload of cane stem and 10 numbers of cane shoots to meet their domestic purposes. Thus, in general the group have managed the NWFP area and resources are well protected as per the management plan prescription.

The group harvested and sold cane stems/culms, cane shoot, *Rubia cordifolia*, mushroom, ferns, *Elatosteme* and *Piper hamiltonii* (pani). In total the group earned income of **Nu.333,050/- three lakh thirty-three thousand fifty** only through sale of these products. Currently the group have a sum of **Nu.45,897/- (forty-five thousand eight hundred ninety-seven)** only as a saving in their group account maintained with Bhutan Development Bank Limited. The detail list of the NWFPs sold and income generated is as given in table below:

Table 2. Income generated through collection and sale of NWFPs

Sl. No	Species harvested	Qty harvested	Qty sold	Amount (Nu.)	Remarks
1	<i>Rubia cordifolia</i>	3870 kgs	3870 kgs	209950	
2	Cane stem/culm	527 bl	527 BL	58100	
3	Cane shoot	690 pcs	690 pcs	34800	
4	Yula	15 bl	15 Bl	450	
5	Mushrooms	100 kgs	100kgs	15000	
6	Piper (Pani)	500 bundle	500 bundle	7500	
7	<i>Diplazium</i> (ferns)	550 bundle	550 bundle	2750	
8	<i>Elatosteme</i> (damroo)	150 bundle	150 bundle	4500	
	Total			333050	

Besides, the artisans from the group made various handicraft products from cane and bamboo such as zem (small, medium and large), basket, lakchung, snacks container, palang, hand basket, serving tray, laptop bag, paper container, baikur, winnower, utensil container, hand bag, tshok offering plates, etc and generated cash income of **Nu.164550/- (one lakh sixty-four thousand five hundred fifty)** only.

From the generated income, the group has utilized about **Nu.86,750/-** for their religious purpose (nungnye) and the group members need not contribute individually. As mentioned by some of the group members the NWFPs benefitted them a lot as it is one of the main sources of income for those household members who don't have alternative source of income generation. The income generated from sale of NWFPs, were utilized for meeting their children's school expenses and to buy household items, pay taxes, etc.

2. Goals and objectives:

- To ensure a sustainable collection of locally available NWFPs through a better management system.
- Income generation through the sale of NWFPs and marketing
- Protection and conservation of NWFPs.

3. Description of NWFPs:

The forest of Broksar has a rich varieties of NWFPs such as *Neomicrocalamus andropogonifolius* (Ringshu), damroo, canes (Phashi, Menji, Drey and yula), *Rubia* (tsut), banana flowers (Pakpali), mushrooms (Muu), *Elatosteme* (Damru), *Diplazium* (Zeelema), *Paris polyphylla* (Thoksumpa), thormangyu (Thormom), Bamboo spp (Rashu/Rawa/Glem/Daii), *Zanthoxylum spp.* Broksar area is mostly dominated by *Plectocomia himalayana* and *Calamus acanthospathus*. They are abundantly found in areas like Thonphu and widely distributed in areas like Shayrila, Yarila, Trangla, Ribalingbu, etc. *Calamus latifolius* and Ringshu are mostly found below the farm road. Therefore, the group will mainly focus on canes (*Plectocomia himalayana*), *Calamus acanthospathus*) and *Rubia cordifolia* (tsut) as the main products for commercial marketing where *Calamus latifolius* (Khran) and *Neomicrocalamus andropogonifolius* (Ringshu) will be collected only if there is a demand from the buyers/traders. NWFPs such as ferns, mushroom, thormangyu, banana flowers, etc will be collected for domestic purposes and for commercial purpose as per the demand.

Table 3. Brief description of NWFPs for commercial harvesting

NWFP name	Harvesting area(s) (name)	Parts harvested	Use purpose (traditional & others)	Harvesting season (months)	Estimated quantity that can be collected	Unit
<i>Plectocomia himalayana</i> (Rey/ Phashi)	Thonphu, Trangla, Yarila, Shayrila, Ribalingbu, Kumaipong	Young shoot as well as matured stem	Young shoots are used as a vegetable. Matured stem used to produce handicraft items	Mature stems- November to March. From shooting April to August.	As per the demand from buyers but not exceeding the AHL.	Backloads
<i>Calamus acanthospathus</i> (Krath/Minji)	Thonphu, Trangla, Yarila, Shayrila, Ribalingbu	Cane	Handicraft	November to March	As per the demand from buyers but not exceeding the AHL	Backloads

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<i>Calamus latifolius</i> (Drey/ Khran)	Ribalingbu, Thonphujug, Thungtong, Chubarchu	Cane	Handicraft	November to March	As per the demand from buyers but not exceeding the AHL	Backloads
<i>Rubia cordifolia</i> (tsod)	Nearby village	Entire plant	Dyes, medicine & rituals	November to February	1500	kg
<i>Neomicrocalamus andropogonifolius</i> (Yula/ Ringshoo)	Khe-khar- tey, Zangzabrang , Kader- sersa, Chubarjug	Culms	Bangchungs and other handicraft items	December to April	1000	Backloads
<i>Chimonobambusa callosa</i> (Rawa)	Thonphu, Throsengbra ng, Rawabee, Shahwabada b, Trong- memba, Phosengbran g,	Culms	Flagpoles, fencing, roofing, some handicrafts	December to February	1000	Nos
<i>Borinda grossa</i> (Rhue)	Broksarphu	Culms	Mats and other products	December to February	1000	Nos
<i>Cepalstachyum</i> (Glim)	Broksarphu, Thonphu, Broksar	Culms	Flagpoles, fencing, roofing, some handicrafts	December to February	3000	Nos
<i>Girardinia diversifolia</i> (Ba-kue/ Zangroo)	Pangkang, Broksar, Korkang, Khongdangb rang, Lungtenbran g, Sharthongsa	Fibre	Weaving	September to October	500	kg
<i>Diplazium esculentum</i> (Zeelema/Nakey)	Pangkang, Gangsengma , Kader-sersa	Young shoots	Vegetable	March to September	1000	bundles
<i>Juglans regia</i> (Walnut)	Gangsengma , Kader- sersa,	Fruit/Nut	Carnal	September to October	1000	kg

	Pangkang, Broksar, Thoipong					
<i>Elatosteme lineolatum</i> (Damroo)	Kader-sersa, Broksar,	Stem and leaves	Vegetable	Throughout the year	200	bundles
<i>Pouzolzia sanguinea</i> (Nagglabma/ Yongjiba)	Khongdangb rang,	Bark	Adhesive, Incense	August to February	200	kg
<i>Piper hamiltonii</i> (Paan)	Khe-khar-tey, Khashalanan g, Kader-sersa,	Leaf	Stimulant	Throughout the year	1000	bundles
<i>Acorus calamus</i> (Chutalee/bar-tsi)	Tshoipong, Broksar	Rhizome	Medicine	October to March	500	Kg
<i>Piper mullesea</i> (Raktang/Pilpa)	Trangla, Trongmanpa, Zabar	Fruit	Medicine, sometimes used as spices	September to November	500	Kg
<i>Dhai</i>	Khe-khar-tey				1000	Nos
<i>Docynia indica</i> (tong)	Pangkang, Thonphu,	Fruit	Candy/jam	December to February	5000	Kg
Homestead Bamboos						
<i>Bambusa nuttan</i>	Nearby village	Culms	House construction	November to February	9000	Nos
<i>Dendrocalamus hamiltonii</i> (Sue)	Nearby village	Culm, young shoots	Weaving, construction materials, etc. Young shoots edible	For culms: November to Feb. For shoots: June to September	1000	Nos
<i>Dendrocalamus sikkimensis</i>	Nearby village	Matured culms	Crafts, house construction	September	200	Nos

4. Harvesting prescriptions for sustainable NWFP harvesting

4.1. Domestic purpose

- **Group Harvesting of Cane**
 - Cane harvesting for domestic use should be done collectively in groups.

- This activity is permitted during any season.
- **Cane Shoot Harvesting Limit:** Each household is allowed to harvest a maximum of 10 cane shoots per season for domestic consumption.
- **Notification Requirement:** Individuals planning to harvest NWFPs for domestic use must notify the Chairperson before beginning the collection.
- **Harvesting Other NWFPs:** NWFPs such as mushrooms, *damroo*, and ferns can be harvested at any time, according to the needs of individual members.
- **Special Circumstances:** In cases of death or emergency, there are no restrictions on harvesting NWFPs.

4.2. Commercial purpose

- **Group Harvesting:** Commercial harvesting must be conducted collectively, involving all members.
- **Permits for Specific NWFPs:**
 - *Mushrooms, Damroo, and Ferns:* Commercial harvesting requires a collection permit from the nearby forest office.
 - *Rubia cordifolia:* Harvesting must occur in the same area and at the same time, following notification from the Chairperson.
 - *Cane Extraction*
 - Only members are allowed to commercially extract cane.
 - Outsiders are strictly prohibited from commercial extraction.
- **Other NWFPs:** Members intending to harvest for commercial or domestic purposes must inform the Chairperson. The Chairperson will facilitate obtaining the necessary collection permit.

The above NWFPs listed in Table 1 should be harvested as per the harvesting prescription shown below in Table 2.

Table 4. Harvesting prescription for sustainable harvesting of NWFPs

NWFP species	Harvesting prescriptions for sustainable harvesting
<i>Rubia cordifolia</i> (Tsod)	● Harvest stems of <i>Rubia</i> after seed dispersal, is mainly from February to April but might start as early as November in the south of Bhutan, depending on the climate and altitude ● Use a sharp sickle or knife for the collection or harvesting of the upper part of the plant. ● Cut the plant at 5 cm above ground level. ● Leave at least 10% of the plants without collecting or harvesting any raw materials from them (these plants shall act as seed plants for natural regeneration). ● Do not collect roots. ● Do not uproot entire plants.
<i>Plectocomia himalayana</i> (Ray), <i>Calamus</i>	For mature stems ● Cut cane stem only when mature, to be recognized by:

<i>acanthospathus</i> (Krath) and <i>Calamus latifolius</i> (Khran)	- Loosening of leaf sheath from the lower part of the cane stem - Exposure of the lowest part of the stem - Turning to yellow-reddish along the entire stem length - The diameter over the entire length of the cane is the same (the diameter at the lower part of the cane is reduced) - Pull down as much of the stem as possible (avoid leaving cane parts high up in the tree) - Cut only canes with a usable length of at least 10 m. - Discard at a maximum of 1.2 to 1.5 meters from the top end. - Harvest during winter as canes have a better quality during this season (lower water content and more durable) For cane shoots - Cut young shoot by bending the cane stem with the help of a long and strong stick with a hook - Do not cut the entire cane for the sake of collecting shoots - Cut the leading shoot with a sharp knife diagonally (not straight to avoid rotting of the remaining stump); leave lateral shoots - Care shall be taken to avoid damaging sucker buds (these will later develop into lateral shoots) Guidelines for both mature stems and cane shoots - Uprooting, cutting of rhizomes/suckers shall be avoided. - Trees supporting the canes shall not be felled - The debris after harvesting canes should be taken outside the cane area to avoid infection with insect borers or fungi
<i>Neomicrocalamus andropogonifolius</i> (Yula/Ringshoo)	- Harvest only culm from a clump that has more than five culms. - Harvest culm when it produces a secondary shoot at the top. - Retain at least two to three young healthy culms in each clump to favour more shooting in the following season.
<i>Chimonobambusa callosa</i> (Rawa); <i>Borinda grossa</i> (Rhue) & <i>Glim</i>	- Cut the culm above the first internode from the base. - Leave bamboo shoots or culm of less than one year old intact. - Cut at a maximum of 33% of the 1+ year and 2+ year old culm per clump. - Do not cut culm when new shoots are sprouting.
<i>Girardinia diversifolia</i> (Ba-kue/Zangroo)	- Peel out fibre after cutting the plant from the base. - Harvest plant only after seed dispersal. - Do not collect all plants from one area.
<i>Diplazium esculentum</i> (Zeelema/Nakey)	- Harvests young shoots that are 15cm in length. - Don't harvest all shoots coming up in the season. - Retain at least 50% of the young shoot of the last flush for the development of leaves. - Don't harvest the shoots that have developed into leaves.

<i>Juglans regia</i> (Walnut)	● Hand-pick fruits only when fallen on the ground, or ● Use long bamboo poles to shake off fruits from the trees. ● Don't cut trees or branches to collect fruits.
<i>Elatosteme lineolatum</i> (Damroo)	● Collect plants that are 15cm and above. ● Retain stump of around 3" for sprouting of new shoots. ● Don't uproot the whole plant. ● Shouldn't cut down trees and create more openings. ● Shouldn't disturb its catchment area.
<i>Pouzolzia sanguinea</i> (Nagglabma/Yongjiba)	● Cut the plant at 10 cm above the ground level. ● Do not damage the remaining young stands/ seedlings/ saplings. ● Should not cut down surrounding trees to ease harvesting. ● Leave at least 10% of the plants to allow natural regeneration. ● Harvesting of roots is strictly prohibited for ecological reasons
<i>Piper hamiltonii</i> (Paan)	● Collect leaves only by hand picking without damaging the climber and its tender buds. ● Harvesting of leaves from a single (same) climber shall be allowed a maximum of 4 times per year. ● Collect a maximum of 80% of the leaves from a single climber. ● Do not cut down the host tree to ease harvesting.
<i>Acorus calamus</i> (Chutalee/bar-tsi)	● Dig up only mature rhizomes without uprooting the whole plant ● Leave the upper part plant intact with a small part of the rhizomes in the soil for regeneration in the following season.
<i>Piper mullesea</i> (Raktang/Pilpa)	● Collect fruits only when fully mature. ● Harvest fruits only after the announced starting date for harvesting (The date is usually announced by the coordinator in September). ● Harvest fruits only by hand picking without damaging, cutting or uprooting the entire plant
<i>Bambusa nutans</i> , <i>Dendrocalamus hamiltonii</i> & <i>D. sikkimensis</i> (Homestead bamboo)	For culms: ● Cut culms above one node from the ground level ● Do not cut culms less than one year old ● Retain 50% of healthy culms in a clump ● Remove matured culms from the centre of the clump. ● Do not harvest the culms growing on the periphery of the clump. For shoots: ● Collect bamboo shoots to a maximum of 25% from a clump ● Cut bamboo shoots with a sharp knife at the ground level without damaging or wounding the other healthy ones ● After cutting and collecting the bamboo shoot, cover that void place with topsoil

Protect the remaining shoots and let them further develop into culms

5. Map of the NWFP harvesting areas

BROKSAR NWFPs MANAGEMENT MAP

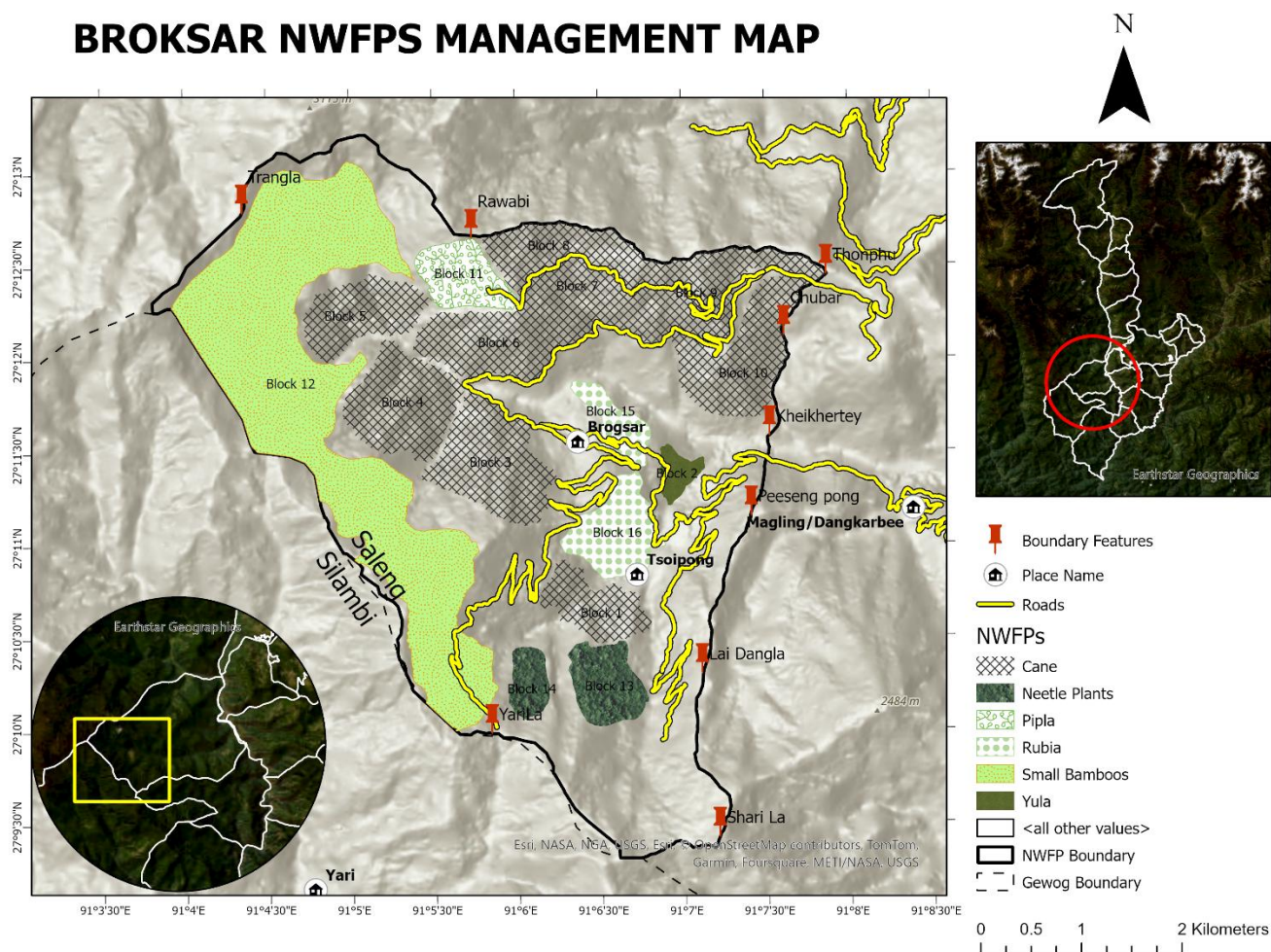


Figure 1: Map of Broksar NWFPs Blocks

6. Marketing and sales plan

- Standard-sized canes, including *Plectocomia himalayana* and *Calamus acanthospathus*, will be sold for commercial handicrafts at a rate of Nu. 10/= per stem, with each cane measuring one meter in length.
- Cane shoots, approximately one meter long, will be sold at a rate of Nu. 70/= per shoot.
- Yula will be sold at Nu. 200/= per backload, with one back load consisting of four bundles, totaling 200 yula stems.
- Group members must pay the permit fee and royalty for NWFPs in terms of backloads.

Note:

- 1 BL of cane stem = 15m length variable sizes
- 1 BL of cane shoot = 15 numbers = 15m length variable size
- 1 BL = 4 bundles = 200 yula stems (1 bundle = 50 pieces)

The marketing and sales plan for different types of NWFPs that are to be collected by the group, its potential markets where the group shall sell products with approximate quantity, price per unit and the total value of the expected sales are described in the below table:

Table 5. Marketing and sales plan of the group

NWFP	Potential market/buyer	Estd. qty for sale per season	Unit	Form of products (Dry/fresh)	Expected price per	Total (Nu)	Remarks
Cane (<i>Calamus acanthospathus</i>)	Within Bhutan	500	Backload	Fresh	10/=	5000	7500 m
Cane (<i>Plectocomia himalayana</i>)	Within Bhutan	500	Backload	Fresh	10/=	5000	7500 m
Cane shoot (<i>Plectocomia himalayana</i> and <i>Calamus acanthospathus</i>)	Within Bhutan	100	Backload	Fresh	70/=	105000	1500 m or pics
<i>Calamus latifolius</i> (Drey/Khran)	Within Bhutan	100	Backload	Fresh	10/=	15000	
<i>Rubia cordifolia</i> (tsod)	India	1500	kg	Dried	80/=	120000	
Yula/Ringshoo (<i>Neomicrocalamus andropogonifolius</i>)	Within Bhutan	125	Backload	Fresh	50/=	25000	500 bundles
Rawa (<i>Chimonobambusa callosa</i>)	Within Bhutan	1000	pieces	Fresh	20/=	20000	
Rhue (<i>Borinda grossa</i>)	Within Bhutan	2000	pieces	Fresh	25/=	50000	
Glim (<i>Cepalstachyum</i> sp)	Within Bhutan	10000	pieces	Fresh	15/=	150000	
Ba-kue/Zangroo (<i>Girardinia diversifolia</i>)	Within Bhutan	100	kg	Fresh	2000/=	200000	
Zeelema/Nakey (<i>Diplazium esculentum</i>)	Within Bhutan	1500	bundles	Fresh	300/=	450000	
Mushroom (Senglengmo)	Within Bhutan	50	kg	Dried	1500/=	75000	
Walnut (Nut) (<i>Juglans regia</i>)	Within Bhutan	1000	kg	dried	150/=	150000	
Damroo (<i>Elatosteme lineolatum</i>)	Within Bhutan	200	bundles	Fresh	20/=	4000	
Nagglabma/Yongjiba (<i>Pouzolzia sanguinea</i>)	Within Bhutan	800	kg	Dried	80/=	64000	
Paan (<i>Piper hamiltonii</i>)	Within Bhutan	200	bundles	Fresh	20/=	4000	
Mushrooms	Within Bhutan	100	kg	Fresh	350/=	35000	
Chutalee/bar-tsi (<i>Acorus calamus</i>)	Within Bhutan	200	kg	Fresh	50/=	10000	

Pilpa (<i>Piper longum/ P. pedicelatum</i>)	Trangla, Trongmanpa, Zabar	200	kg	Dried	150/=	30000	
Thogsumpa (<i>Paris pollyphylla</i>)	Within Bhutan	150	kg	Dried	1500/=	225000	
Dhai	Within Bhutan	300	Nos	Fresh	20/=	6000	
Homestead bamboo							
<i>Bambusa nuttan</i>	Within Bhutan	9000	Nos	Fresh	150/=	1350000	
<i>Dendrocalamus hamiltonii</i> (Sue)	Within Bhutan	1000	Nos	Fresh	150/=	150000	
<i>Dendrocalamus sikkimensis</i>	Within Bhutan	200	Nos	Fresh	150/=	30000	

7. Marketing arrangements

The Chairperson, with the support of the nearest forest office, shall assess and gather the demand for NWFPs before the onset of any product collection season. The marketing of NWFPs will then be organized and arranged according to the following procedures in Table 4:

Table 6. Procedure for NWFP harvesting, marketing and sharing of income

Activity	A brief description of how the activities are carried out
<i>Harvesting of products</i>	NWFPs shall be harvested and marketed as a group. Collection should occur simultaneously on the same date and time, as directed by the Chairperson. All members should participate at the same collection site for NWFP harvesting.
<i>Post-harvest practices</i>	- After harvesting, NWFPs should be cleaned, sorted, and processed according to the standards set by the group to ensure quality and market readiness. - Proper storage facilities should be used to maintain the quality of NWFPs until they are marketed. This includes protecting the products from pests, moisture, and other deteriorating factors. - Packaging should be done in a manner that preserves the quality and facilitates easy transport and handling
<i>Marketing of products</i>	- All marketing activities will be coordinated by the Chairperson, ensuring that members adhere to the agreed-upon schedule and procedures. - Members will deliver their products to the designated location on the date agreed upon with the buyer. - Whenever feasible, an open auction will be organized to ensure transparent pricing and fair competition among buyers. - All transactions will be documented, including buyer details, quantities sold, and prices agreed upon, to maintain transparency and accountability. - Regular feedback from buyers will be sought to improve product quality and address any market demands or concerns
<i>Identification of buyers</i>	- Buyers must hold a valid license for purchasing NWFPs. - Preference should be given to buyers with a strong reputation for fair trade practices and timely payments. - Buyers should have the capacity to purchase large volumes, ensuring a stable and continuous market for the products. - Buyers should demonstrate a commitment to sustainable and ethical sourcing practices.

<i>Price fixing</i>	- Buyers should offer competitive prices that reflect the market value of the NWFPs. - Buyers must agree to the payment terms set by the group, including no advance payments and full payment upon delivery - Members are not allowed to accept advance payments from buyers. However, they have the right to sell to any licensed buyer who offers a better price to the group
<i>Income sharing</i>	60% of the income from the sale of the NWFPs shall be shared among the group members no matter how much quantity of the products individual members have collected and 40 % will be deposited in the group bank account.
<i>Raising of community development fund</i>	The 40 % of the total income generated shall be deposited in the group savings account
<i>Payment to committee members for their services</i>	Executive members shall receive an incentive of Nu. 150/- per occurrence for each sale of NWFPs conducted within the village (<i>refer to By-laws, section 13</i>).

8. Benefits to the members

- All group members shall have equal rights to collect NWFPs and access to the market.
- 60% of the income from NWFP sales will be distributed equally among all group members, regardless of the quantity each member has collected. The remaining 40% will be deposited into the group bank account.
- Members who are absent during the collection time will not receive a share of that day's income.

Table 7. Financial projection and benefits to the members from sale of NWFPs

Sl.#	Particulars	Product (names)					Total amount
		Cane (BL)	Cane shoot (Nos.)	Yula (BL)	Rubia (kg)	Pipla (kg)	
1	Revenues						
a	Total quantity (unit)	1,100.00	1,000.00	125.00	1,500.00	200.00	
b	Market price (Nu per unit)	10.00	70.00	200.00	80.00	150.00	
c	Total revenues from forest product sales (a*b)	11,000.00	70,000.00	25,000.00	120,000.00	30,000.00	256,000.00
A	Total revenues	11,000.00	70,000.00	25,000.00	120,000.00	30,000.00	226,000.00
2	Expenses						
a	Packaging materials	0	0		0		0
b	Royalty	4,400.00	4,000.00	125.00	24,000.00	4,000.00	36525
c	Transportation costs	0	0	0	0	0	0
d	Compensation fee to the executive committee for marketing products	300	300	300	300	300	1500

e	Contribution to community development fund	4,400.00	28,000.00	10,000.00	48,000.00	12,000.00	90,400.00
f	Incidental expenses	0	0	0	0	0	0
B	Total expenses	9100	32300	10425	72300	16300	354,425.00
C	Total benefit (net income) of group members (total revenues A – total expenses B)	1,900.00	37,700.00	14,575.00	47,700.00	13,700.00	115,575.00
D	Average income to group members (C/ no of group members)	79.17	1,570.83	607.29	1,987.50	570.83	4,070.06
E	Price per unit paid to group members (C/1a)	1.73	37.70	116.60	31.80	68.50	

9. Benefits to the group

- Group members shall have equal rights to collect and market NWFPs.
- The group will benefit from membership fees and 40% of the sale value deducted from individual members based on the quantity they collect and sell.
- Develop a sense of ownership over the resources and provide their support and commitments in the conservation, protection, and sustainability of resources.

Table 8. Financial projection of the benefits to the group, the community development fund

Sl. #	Particulars	Amount (Nu)
	Revenues	
1	Membership fees	1,200.00
2	Contribution to the Community Development Fund (CDF) from sales of product 1 (Cane)	4,400.00
3	Contribution to the Community Development Fund (CDF) from sales of product 2 (Cane shoot)	28,000.00
4	Contribution to the Community Development Fund (CDF) from sales of product 3 (Yula)	10,000.00
5	Contribution to the Community Development Fund (CDF) from sales of product 4 (Rubia)	48,000.00
6	Contribution to the Community Development Fund (CDF) from sales of product 5 (Pipla)	12,000.00
7	Fines for absentees in group meeting	-
A	Total revenues	103,600.00

	Expenses (only include expenses that will be paid from the group funds)	
1	Stationeries	1,500.00
2	Working lunch & Refreshments during meetings and visitors	2,400.00
3	Compensation fee to the executive committee for marketing products	1,500.00
4	Transportation (Lumpsum)	5,000.00
B	Total Expenses	10,400.00
C	Balance: Total revenues – total expenses	93,200.00

10. Cane Resource Assessment

The resource assessment for cane species (*Plectocomia himalayana* and *Calamus acanthospathus*) was conducted in the Thonphu area. The assessment involved the collection of data from respective locations and quantities of cane species within a specific region. Based on the survey results, the area was divided into 11 blocks according to the availability of resources, using Google Maps and participatory methods. Among these blocks, farmers agreed to focus on collecting cane (*Plectocomia himalayana* and *Calamus acanthospathus*) from Thonphu, as it has the highest concentration of these species. Other cane species, such as *Calamus latifolius*, along with some *Plectocomia himalayana* and *Calamus acanthospathus*, were also found in other blocks. Additionally, there is one block designated for bamboo (Yula), and the homestead bamboo were assessed, showing a promising stock for both domestic and market purposes.

Table 9. Annual Harvesting Limit (AHL) and annual demand status of Cane and Bamboo

Species/Product	Total Annual Demand	Annual Harvesting Limit	Variation	Remarks
<i>Calamus acanthospathus</i>	300 backloads	580 backloads	+280 backloads	Surplus/deficit
<i>Plectocomia himalayana</i>	500 backloads	5072 backloads	+4572 backloads	
Yula	100 backloads	270.49 backload	+170.49 backloads	It has flowered and started dyeing
<i>Bambusa nuttans</i>	Nil	810 numbers	+810 culms of pieces	From private registered land
<i>Dendrocalamus hamiltonii</i> (Sue)	450 numbers	953 numbers	+503 culms of pieces	
<i>Dendrocalamus sikkimensis</i>	Nil	60 numbers	+60 pieces of culms	

The resource assessment for cane species was specifically conducted in Blocks 1 and 3 to 10, while Block 2 was assessed for Yula, following the guidelines for resource assessment and management of canes developed by DoFPS. The cane species (*Plectocomia himalayana* and *Calamus acanthospathus*) are abundant in these blocks, making them the primary focus areas.

A stratified systematic grid sampling method was adopted for this cane and bamboo assessment. Grids were generated using GIS software (QGIS), with grid distances set at 150m by 150m and circular plots with a radius of 12.62m. The number of sample plots generated is detailed in Table 7.

Table 10. Overview of the cane, bamboo and other NWFPs block area

Name	Area Ha	Produces	Plots (150mX150m)
Block 1	50.01	Cane	33
Block 2	15.51	Yula	10
Block 3	100.80	Cane	67
Block 4	92.22	Cane	61
Block 5	65.93	Cane	44
Block 6	90.21	Cane	60
Block 7	48.95	Cane	33
Block 8	64.57	Cane	43
Block 9	104.46	Cane	70
Block 10	95.73	Cane	64
Other NWFPs area	1804.72	Others	
Total	2533.11		

11. Analysis of Resource Assessment Data

11.1. Cane resources

Table 8 shows the result of over annual harvesting limits for cane harvesting over 10 years in the Boksar NWFP area. The block-wise AHL analysis of cane is given in the Annexure.

Table 11. Annual Harvesting Limit of Canes (All Blocks)

AHL of canes in number of stems and backloads over 10 years											
Period	Cane Species	Blocks									Total
		1	3	4	5	6	7	8	9	10	
Year 1-2: Present estimated length: $\geq 15\text{m}$	<i>Calamus acanthospathus</i> (Krath)	8	0	8	0	4	142	0	49	109	320
	<i>Plectocomia himalayana</i> (Patsha)	69	40	36	12	384	1064	32	817	202	2659
Year 3-4: Present estimated length: 10-14.9m	<i>Calamus acanthospathus</i> (Krath)	0	0	0	0	5	0	0	51	32	87
	<i>Plectocomia himalayana</i> (Patsha)	19	27	27	0	221	265	46	148	102	855
Year 5-6: Present estimated length 6 -9.9m	<i>Calamus acanthospathus</i> (Krath)	0	0	0	0	11	9	0	15	11	46
	<i>Plectocomia himalayana</i> (Patsha)	0	6	9	2	190	127	6	131	8	478
Year 7-8: Present estimated length 3 - 5.9m	<i>Calamus acanthospathus</i> (Krath)	11	0	0	0	9	11	0	0	16	48
	<i>Plectocomia himalayana</i> (Patsha)	3	5	11	0	80	119	5	201	44	468
	<i>Calamus acanthospathus</i> (Krath)	0	0	0	0	31	10	0	20	18	80

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Year 9-10: Present estimated length 1-2.9m	<i>Plectocomia himalayana</i> (Patsha)	18	5	7	1	138	174	13	203	52	612
Total	<i>Calamus acanthospathus</i> (Krath)	19	0	8	0	60	172	0	135	186	580
	<i>Plectocomia himalayana</i> (Patsha)	109	83	91	14	1013	1750	103	1500	408	5072

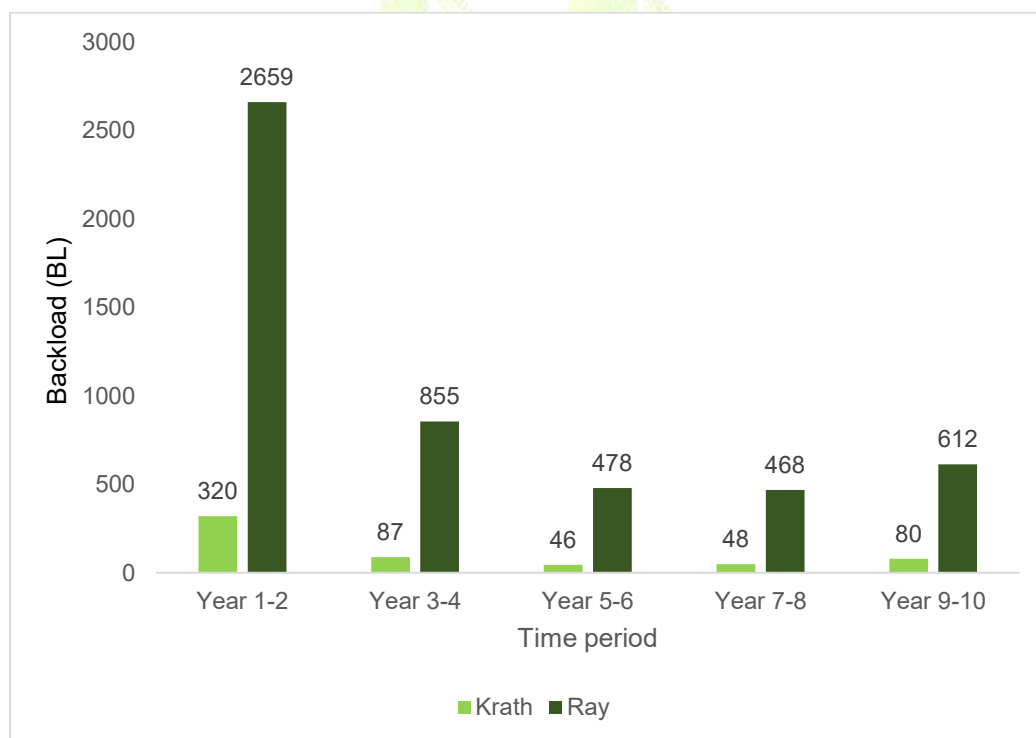


Figure 2: AHL of canes for 10-year period

11.2. Yula Resources

Table 12. Annual Harvesting Limit of Yula

Sub-Block		Sample plots		No of Yula clums to be harvested this year			No. of Yula clums harvested last year			Average no. of culms per backload [i]	Range of AHL of Yula backloads [j]=(e/i - h/i)
Name	Area (Ha)[a]	No s	Total area (Ha)[b]	in plots [c]	per hectare [d]=(c/b)	in sub-blocks [e]=(d*a)	in plots[f]	Per acre [g]=(f/b)	in sub-blocks [h]=(g*a)		
B2	15.51	20	1.00	104	104	1613	70	70	1086	2	270
Total											270.49

PART B: GROUP'S BY-LAWS

10. NWFP management and marketing group	
Membership arrangement and resignation	Membership Arrangement • Eligibility for Membership: All residents of Broksar Village are eligible for membership upon payment of the cumulative annual membership fee and contribution of labour. • Use of Resources by Non-Members: Individuals who choose not to become members are prohibited from using these resources for commercial purposes. However, they may use these resources for personal, non-commercial purposes, provided they have consulted with the Executive Members. • Regulation of Non-Compliance: Any household found using Non-Wood Forest Products (NWFPs) for commercial purposes without proper membership will be subject to enforcement actions per the existing rules and regulations. Procedure for withdrawal • Voluntary Withdrawal: Any member wishing to withdraw from the group may do so freely. However, upon voluntary withdrawal, the member forfeits any entitlement to forest resources. Additionally, any fees paid or labour contributions made during their membership will not be refunded. • Involuntary Withdrawal Due to Disadvantaged Circumstances: Members who must withdraw involuntarily due to disadvantaged circumstances, such as being a single-family member, may exit the group. In these cases, the withdrawing member will be entitled to receive an equal share of the total accumulated funds of the group.
	Responsibilities and rights of members: • Sustainable Management: All members are responsible for the sustainable management of NWFPs, ensuring adherence to the principles and practices outlined in the management plan. • Adherence to Harvesting Seasons: The harvesting of NWFPs must comply with the designated seasons specified in the management plan. • Compliance with Executive Instructions: Members shall follow and respect the instructions issued by the executive members regarding the management and harvesting of NWFPs. • Vigilance Against Illegal Activities: Members are obligated to remain vigilant against the illegal harvesting of NWFPs by non-members. Any such activities must be reported promptly to the forest office or executive members. • Rights to Harvest and Market: All group members have the right to harvest and market NWFPs per the management plan.

11. Executive committee of NWFP management and marketing group

Chairperson	Name of Chairperson: Mr. Tagyal (Contact No. 17977480) Responsibilities of Chairperson: • Meeting Leadership and Dispute Resolution: Lead meetings and resolve disputes and conflicts among members. • Coordination of Group Management Tasks: Coordinate group management tasks, including contacting traders, organizing, and exploring the marketing of NWFPs. • Enforcement of Fines and Penalties: Enforce fines and penalties following the bylaws. • Permit Application: Apply for NWFP collection permits from the forest office during harvesting seasons. • Notification of Collection Dates: Notify each member of the NWFP collection dates. • Coordination of Harvesting Activities: Coordinate NWFP harvesting activities. • Participation in Training and Development: Attend training sessions, workshops, or study tours as required.
Secretary/ Treasurer	Name of Treasurer: Pelzom Buthri (Contact No. 17701780) Responsibilities of Treasurer/ Secretary: • Financial Recordkeeping: The Treasurer/Secretary shall maintain accurate records of all cash transactions. • Collection and Deposit of Fees: The Treasurer/Secretary is responsible for collecting membership fees, NWFP fees, and deposits, and ensuring these are deposited in the designated bank account. • Sales Record Maintenance: The Treasurer/Secretary must maintain detailed records of NWFP sales data. • Financial Reporting: The Treasurer/Secretary shall present the group's annual expenditure and income to the members during meetings. • Sales Records: The Treasurer/Secretary shall maintain comprehensive records of all sales transactions. • Minutes of Meetings: The Treasurer/Secretary is required to maintain accurate minutes of all meetings. • Training and Participation: The Treasurer/Secretary shall attend and participate in training, workshops, and study tours when called upon. • Support to the Chairperson: The Treasurer/Secretary shall assist the Chairperson as needed. • Harvest Records: The Treasurer/Secretary must keep detailed records of NWFP harvests and submit these records to the Chairperson.

12. Tenure of committee member

- The appointed committee shall serve a term of five years. However, if a change in the committee is deemed necessary, the decision shall be made by the group during a duly convened meeting.

13. Incentives to the committee members

Incentive for NWFP Sales:

- Executive members shall receive an incentive of Nu. 150/- per occurrence for each sale of NWFPs conducted within the village.

Daily Allowance and Reimbursement for Meetings and Training:

- If travel expenses are not covered by the government or other organizations during meetings and training sessions, the group shall provide a daily allowance of Nu. 500/- per day to executive members.
- Additionally, food expenses will be reimbursed based on the submitted cash memo.
- Travel costs incurred by executive members during the program shall be borne by the members themselves, to be settled after the conclusion of the program.

14. Sources of generation of group fund

Membership registration fees	• Every member is required to contribute Nu. 50/- annually as a membership fee during the annual meeting throughout the entire plan period. • The membership fees collected from group members shall be deposited in the community development fund in any bank.
Collection from the sale of NWFP produce/ products	• An amount equivalent to 40% of the gross sales, inclusive of royalty fees, shall be deposited into the group account quarterly.
Fines and Penalties	• Fines and penalties shall be imposed for violations of the By-laws and non-compliance with the management plan as outlined in the following section.
Other (not listed)	• The funds will be accrued from sources not listed above, including donations and contributions from visitors and donors.

15. Group fund management and utilization

- **Deposits:** All cash collected from various sources shall be deposited into the group account maintained with BDBL.
- **Loans to Members:** Loans to group members shall be provided at an interest rate of 6% per annum.
- **Utilization of Funds:** The funds shall be utilized for the following purposes:
 - Community religious rituals
 - Maintenance of the community market work shed
 - Incentives for the committee
 - Repair and maintenance of community tools and equipment
 - Procurement of tools and equipment
 - Remuneration for meetings and refreshments
 - NWFP resource development
- **Support for Affected Families:** The group shall support families affected by natural calamities and domestic problems, providing assistance amounting to Nu. 3,000 per incident.

16. Conflict Resolution

- **Internal Conflicts:** Any conflicts or disagreements arising within the NWFP Management Group that cannot be resolved internally by its members shall be escalated to the Chairperson for resolution.

- **Referral to Gewog Administration:** If the Chairperson is unable to resolve the issue, it shall be referred to the Gewog Administration through the Chiwog Tshogpa for further resolution.
- **Conflicts with External Parties:** In the event of a conflict between the NWFP Group and external parties, the Chairperson shall forward the case to both the Divisional Forest Office and the Gewog Administration for appropriate action.

17. Meeting and purpose

- **Annual Meeting:** An annual meeting shall be held once a year to address matters concerning progress evaluation, and strategic planning, and to seek approval for the collection and marketing of NWFPs.
- **Special Meetings:** Additional meetings may be convened as needed to discuss urgent matters, conduct training sessions, or address other pertinent issues requiring immediate attention.

18. Penalties

- **Access to Resources:** Members who leave the group shall forfeit all access to the group's resources.
- **Non-Payment of Fees:** Members who fail to pay their annual fees will be removed from the group.
- **Meeting Attendance and Punctuality:**
 - Members who fail to attend meetings shall incur a fine of Nu. 150 per missed meeting.
 - Members arriving more than ten minutes late shall incur a fine of Nu. 50.
- **Work Absence:** Members who are absent from scheduled work shall incur a fine of Nu. 500.
- **Unauthorized Collection of NWFPs:** The collection of non-listed NWFPs shall be subject to fines as per Rule 338 of the FNCRR, 2023.

19. Powers of the Department

- **Authority to Terminate:** The Department reserves the right to terminate the NWFP Management and Marketing Group in the event of any breach of the management plan's provisions or failure to execute activities per the plan.
- **Compliance and Enforcement:** According to Section 145 of the FNCA, 2023, any NWFP Management Group that fails to adhere to the NWFP Management Plan may be subject to suspension, reinstatement, or termination per the procedures stipulated in the applicable Rules.

20. Cancellation of NWFP Management Certificate

In accordance with Chapter III, sections 156, 157 & 158 of FNCRR, 2023:

- The NWFP Management Certificate shall be revoked if the performance of NWFMG does not conform to the provisions of the approved management plan or if the sustainability of the resource is jeopardized.
- Should the members of the NWFMG elect to revoke the NWFP Management Certificate according to these Rules, such revocation shall take effect on the date the decision is communicated in writing to the CFO. The NWFMG must then promptly surrender the Certificate to the Department.
- The CFO retains the authority to investigate the grounds for such revocation to ensure compliance with regulatory requirements and shall notify the NWFMG upon the cancellation of the certificate.

Annexure I: NWFP Group Membership List

Sl.#	Name	CID #	Gung No.	Thram No.
1	Ugyen	10711000173	Ja-11-311	288
2	Tshewang Zangmo	10711000151	Ja-11-224	314
3	Tshering Dema	10711000239	Ja-11-234	230
4	Karma	10711000180	Ja-11-228	228
5	Leki Zangmo	10711000083	Ja-11-219	694
6	Tshering Wangmo	10711000130	Ja-11-222	222
7	Dorji Wangmo	10711000093	Ja-11-220	487
8	Dorji Dema	10711000077	Ja-11- 218	231
9	Tashi Zangmo	10711000091	Ja-11-220	487
10	ButhriMo	10711000236	Ja-11-234	230
11	Nima Zangmo	10711000110	Ja-11-221	232
12	Sangay	10711000090	Ja-11-310	487
13	Yeshi Lhamo	10711000175	Ja-11-227	228
14	Nima Chogyal	10711000221	Ja-11-231	144
15	Pema Tenzin	10711000188	Ja-11-229	227
16	Leki Tshering	10711000138	Ja-11-223	222
17	Sangay Zangmo	10711000158	Ja-11-225	313
18	Leki Wangdi	10711000208	Ja-11-230	224
19	Tshewang Lhadon	10711000251	Ja-11- 234	230
20	Sangay Zangmo	10711000092	Ja-11-403	1114
21	Tagyal	10711000296	Ja-11-238	493
22	Pelzang Buthri	10711000187	Ja-11-228	228

Annexure II: Block description form for canes and yula

Date: November 2023

Village: Broksar

Name of cane area: Reflected in Table 7

Block name: Reflected in Table 7

Total area (ha): Reflected in Table 7

Access	<i>What is the walking distance from the nearest village to the block?</i>		
	Less than 3 hours	Within a day	More than a day
	<i>What is the walking distance from the block to the nearest road?</i>		
	Less than 3 hours	Within a day	More than a day

Slope	Very steep >100%	Steep 70-100%	Moderate 45-70%	Gentle < 45 %
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Forest description	Dominant and associate species, canopy cover: <i>Symplocos sp., Toona sp., Quercus sp., Michelia sp., Lithocarpus sp., Persea sp.,</i>
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Sub-block	No.	Density	Area (ha)	No of sample plots
	<i>Reflected in Table 7</i>			

Plantation/ enrichment potential	Sub-blocks	Has plantation potential	No potential	Not applicable
	Block 1 (Cane)		√	
	Block 2 (Yula)		√	
	Block 3 (Cane)			√
	Block 4 (Cane)			√
	Block 5 (Cane)			√
	Block 6 (Cane)			√
	Block 7 (Cane)			√
	Block 8 (Cane)			√
	Block 9 (Cane)			√
	Block 10 (Cane)			√

	Other NWFPs			√
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Current management	Current management practices in entire block: <i>The first management plan was completed and the NWFPs were managed as per the prescribed management plans.</i>
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	Threat	Description
Current threats	Fire	<i>Since the large portion of management area falls in the cool broadleaf there is less chances of forest fire. However, the two bamboo blocks fall in the mixed chirpine and broadleaf forests</i>
	Grazing	<i>There is grazing pressure in the management areas</i>
	Illegal exploitation	<i>There is illegal exploitation of NWFPs by other outsiders like NRDCL logging labours since the NWFPs management area falls in the production circle.</i>
	Over-exploitation	<i>During the resource assessment, there is no sign of over-exploitation of NWFPs</i>
	Other:	

Opportunities	Future opportunities for the entire block: <i>The cane, bamboo and other NWFPs can be harvested for the livelihood improvement of the locality. The proper management would regrowth of the NWFPs.</i>
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Annexure III: Data Compilation Sheet

Block	Plot No	Plectocomia himalayana(Ray/Pastha)					Calamus acanthospathus(krat)					Calamus latifolius/Khran)				
		≥15 m	10-14.9m	6-9.9 m	3-5.9 m	1-2.9 m	≥15 m	10-14.9m	6-9.9 m	3-5.9 m	1-2.9 m	≥15 m	10-14.9m	6-9.9 m	3-5.9 m	1-2.9 m
B1	B1_1	2	0	0	0	1	2	0	0	10	0	0	0	0	0	0
B1	B1_2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B1	B1_3	1	0	0	0	5	0	0	0	1	0	0	0	0	0	0
B1	B1_4	1	1	0	0	4	0	0	0	0	0	0	0	0	0	0
B1	B1_5	1	3	0	0	2	0	0	0	0	0	0	0	0	0	0
B1	B1_6	1	0	0	0	2	0	0	0	0	0	0	0	0	0	0
B1	B1_7	2	0	0	0	7	0	0	0	0	0	0	0	0	0	0
B1	B1_8	2	0	0	0	7	0	0	0	0	0	0	0	0	0	0
B1	B1_9	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B1	B1_10	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B1	B1_11	1	0	0	3	0	0	0	0	0	0	0	0	0	0	0
B1	B1_12	1	4	0	0	2	0	0	0	0	0	0	0	0	0	0
B3	B3_1	0	3	2	2	0	0	0	0	0	0	0	0	0	0	0
B3	B3_2	2	0	0	2	1	0	0	0	0	0	0	0	0	0	0
B3	B3_3	2	4	0	0	2	0	0	0	0	0	0	0	0	0	0
B3	B3_4	4	4	1	1	2	0	0	0	0	0	0	0	0	0	0
B3	B3_5	1	0	0	0	3	0	0	0	0	0	0	0	0	0	0
B3	B3_6	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
B4	B4_1	0	1	1	0	1	1	0	0	0	0	0	0	0	0	0
B4	B4_2	0	1	2	1	1	2	0	0	0	0	0	0	0	0	0
B4	B4_3	0	1	1	2	1	1	0	0	0	0	0	0	0	0	0
B4	B4_4	0	1	3	0	2	2	0	0	0	0	0	0	0	0	0

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B4	B4_5	0	1	2	1	1	2	0	0	0	0	0	0	0	0	0
B4	B4_6	0	2	1	1	2	1	2	0	0	0	0	0	0	0	0
B4	B4_7	0	1	0	1	1	1	0	0	0	0	0	0	0	0	0
B4	B4_8	0	1	1	0	2	2	0	0	0	0	0	0	0	0	0
B5	B5_1	6	0	3	0	1	0	1	0	0	0	0	0	0	0	0
B6	B6_10	0	0	0	0	0	0	0	0	0	0	0	0	18	0	0
B6	B6_11	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0
B6	B6_12	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0
B6	B6_13	0	1	0	0	0	0	0	0	0	0	0	6	4	0	6
B6	B6_14	0	0	0	0	0	0	0	0	0	0	0	7	5	10	15
B6	B6_16	0	0	0	0	0	0	0	0	0	0	0	10	15	20	43
B6	B6_20	0	0	0	0	0	0	0	0	0	0	0	2	1	0	0
B6	B6_21	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0
B6	B6_22	0	0	0	0	0	0	0	0	0	0	0	0	0	5	0
B6	B6_25	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
B6	B6_26	0	0	0	3	0	0	0	0	0	0	0	6	6	0	8
B6	B6_28	0	0	0	3	0	0	0	0	0	0	0	5	8	10	1
B6	B6_29	0	0	0	0	0	0	0	0	0	0	0	5	15	10	6
B6	B6_30	1	0	0	0	0	0	0	0	0	0	0	6	5	7	4
B6	B6_31	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2
B6	B6_32	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B6	B6_33	0	0	0	0	0	0	0	0	0	0	0	5	0	0	8
B6	B6_34	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0
B6	B6_35	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0
B6	B6_36	1	0	0	0	0	0	0	0	0	0	0	10	6	0	10

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B6	B6_6	0	0	0	0	0	0	0	0	0	0	17	0	3	1	0
B6	B6_7	0	0	0	0	0	0	0	0	0	0	13	16	28	15	0
B6	B6_8	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0
B6	B6_9	0	0	0	0	0	0	0	0	0	0	0	8	3	7	0
B7	B7_10	0	0	0	0	0	0	0	0	0	0	38	0	0	0	0
B7	B7_12	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0
B7	B7_13	0	0	0	0	0	0	0	0	0	0	7	5	5	4	0
B7	B7_14	0	0	3	0	0	0	0	0	0	0	50	30	20	60	0
B7	B7_16	0	0	0	0	0	0	0	0	0	0	38	0	0	0	0
B7	B7_17	0	0	0	0	0	0	0	0	0	0	0	15	2	0	0
B7	B7_18	0	0	0	0	0	0	0	0	0	0	0	0	1	3	0
B7	B7_19	30	0	0	10	0	0	0	0	0	0	15	0	8	0	0
B7	B7_2	0	0	0	0	0	0	0	0	0	0	25	9	0	15	0
B7	B7_20	0	0	3	1	0	0	0	0	0	0	1	0	0	0	0
B7	B7_3	0	0	0	0	0	0	0	0	0	0	0	0	8	1	0
B7	B7_5	0	0	0	0	0	0	0	0	0	0	20	10	8	19	0
B7	B7_6	0	0	0	0	0	0	0	0	0	0	7	6	8	5	0
B7	B7_8	5	0	0	0	0	0	0	0	0	0	45	30	10	10	0
B7	B7_9	0	0	0	0	0	0	0	0	0	0	17	4	4	1	0
B7	B7_16	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
B8	B8_14	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0
B8	B8_24	0	0	0	0	0	0	0	0	0	0	0	10	0	0	0
B8	B8_29	0	0	0	0	0	0	0	0	0	0	5	1	1	1	0
B8	B8_6	0	0	0	0	0	0	0	0	0	0	0	1	2	3	0
B8	B8_8	0	0	0	0	0	0	0	0	0	0	3	2	1	1	0

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B9	B6_10	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0
B9	B6_16	0	0	0	0	0	0	0	0	0	0	14	10	9	9	0
B9	B6_7	0	0	0	0	0	0	0	0	0	0	1	0	0	2	0
B9	B6_8	0	0	0	0	0	0	0	0	0	0	2	0	1	8	0
B9	B6_9	0	0	0	0	0	0	0	0	0	0	50	0	46	20	0
B9	B8_29	0	0	0	0	0	0	0	0	0	0	0	0	5	0	0
B9	B9_12	0	0	0	0	0	0	0	0	0	0	0	0	0	10	0
B9	B9_13	0	0	0	0	0	0	0	0	0	0	4	0	1	4	0
B9	B9_15	0	0	0	0	0	0	0	0	0	0	6	5	3	14	0
B9	B9_20	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0
B9	B9_21	0	0	0	0	0	0	0	0	0	0	5	0	0	15	0
B9	B9_22	0	0	0	0	0	0	0	0	0	0	7	0	10	0	0
B9	B9_24	0	0	0	0	0	0	0	0	0	0	0	4	2	8	0
B9	B9_25	0	0	0	0	0	0	0	0	0	0	12	5	8	15	0
B9	B9_26	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
B9	B9_27	0	0	0	0	0	0	0	0	0	0	13	9	14	9	0
B9	B9_28	0	0	0	0	0	0	0	0	0	0	0	0	0	5	0
B9	B9_29	0	0	0	0	0	0	0	0	0	0	20	3	0	0	0
B9	B9_30	0	0	0	0	0	0	0	0	0	0	1	0	0	5	0
B9	B9_33	7	0	0	0	0	0	0	0	0	0	6	0	0	0	0
B9	B9_35	0	3	0	0	0	0	0	0	0	0	0	0	0	6	0
B9	B9_36	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0
B9	B9_37	0	0	0	0	0	0	0	0	0	0	0	1	0	3	0
B9	B9_38	5	18	0	0	0	0	0	0	0	0	20	0	20	15	0
B9	B9_40	0	0	0	0	0	0	0	0	0	0	0	0	0	6	0

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B9	B9_41	0	0	6	0	0	0	0	0	0	0	0	9	0	0	0
B9	B9_43	0	0	0	0	0	0	0	0	0	0	0	0	0	6	0
B9	B9_5	0	0	0	0	0	0	0	0	0	0	35	15	10	37	0
B10	B10_1	0	2	1	1	0	2	1	1	0	0	0	0	0	0	0
B10	B10_2	0	5	1	0	0	0	0	1	0	1	0	0	0	0	0
B10	B10_3	0	4	2	0	1	0	2	0	0	0	0	0	0	0	0
B10	B10_4	0	1	1	0	1	0	3	1	0	1	1	0	0	0	0
B10	B10_5	0	1	0	0	0	1	0	3	2	1	1	0	0	0	0
B10	B10_6	0	4	0	0	0	0	0	3	0	1	1	0	0	0	0
B10	B10_7	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0
B10	B10_8	0	1	0	0	0	0	2	0	0	0	0	0	0	0	0
B10	B10_9	0	2	7	0	3	20	1	0	0	0	1	0	0	0	0
B10	B10_10	0	1	0	0	0	0	2	3	0		3	0	0	0	0
B10	B10_11	0	4	10	0	5	10	3	0	0	0	0	0	0	0	0
B10	B10_12	0	7	0	0	10	10	2	0	0	0	4	0	0	0	0
B10	B10_13	0	0	6	0	4	4	2	0	0	0	6	0	0	0	0
B10	B10_14	0	4	0	0	10	0	3	0	0	0	4	0	0	0	0
B10	B10_15	0	3	0	0	0	10	0	0	0	4	0	0	0	0	0
B10	B10_16	0	2	8	0	0	8	1	0	0	0	0	0	0	0	0
B10	B10_17	0	5	6	0	0	0	2	0	0	0	9	0	0	0	0
B10	B10_18	3	0	0	0	2	3	0	0	0	1	0	0	0	0	0
B10	B10_19	0	0	0	0	0	3	0	0	1	2	0	0	0	0	0
B10	B10_20	1	1	0	3	2	3	0	0	0	0	0	0	0	0	0

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B10	B10_2 1	1	1	0	0	3	2	0	0	0	0	0	0	0	0	0
B10	B10_2 2	1	0	0	1	1	0	0	0	1	2	0	0	0	0	0
B10	B10_2 3	0	1	0	0	0	3	1	0	1	0	0	0	0	0	0
B10	B10_2 4	1	0	0	0	1	2	0	0	1	2	0	0	0	0	0
B10	B10_2 5	0	1	0	0	0	3	1	0	1	0	0	0	0	0	0
B10	B10_2 6	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0